

Practical – 5

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data, and the child process consumes it. Measure the communication efficiency. Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

1. Producer–Consumer Using Anonymous Pipes

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/time.h>
#include <string.h>

#define NUM_MESSAGES 10000

int main()
{
    int pipefd[2];
    pid_t pid;
    char buffer[100];

    /* Create pipe */
    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    /* Create child process */
    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        exit(1);
    }

    /* Child process - Consumer */
    if (pid == 0)
    {
        close(pipefd[1]);

        int total_bytes = 0;
        int bytes_read;

        while ((bytes_read = read(pipefd[0], buffer, sizeof(buffer))) > 0)
        {
            total_bytes += bytes_read;
        }

        printf("Consumer received %d bytes\n", total_bytes);

        close(pipefd[0]);
    }
}
```

```

/* Parent process - Producer */
else
{
    close(pipefd[0]);

    struct timeval start, end;
    gettimeofday(&start, NULL);

    for (int i = 0; i < NUM_MESSAGES; i++)
    {
        sprintf(buffer, "Message %d\n", i + 1);
        write(pipefd[1], buffer, strlen(buffer));
    }

    close(pipefd[1]);

    wait(NULL);

    gettimeofday(&end, NULL);

    double time_taken =
        (end.tv_sec - start.tv_sec) * 1000000.0 +
        (end.tv_usec - start.tv_usec);

    printf("Producer sent %d messages\n", NUM_MESSAGES);
    printf("Communication time: %.2f microseconds\n", time_taken);

    if (time_taken > 0)
    {
        double throughput =
            (NUM_MESSAGES * 10.0) / time_taken;

        printf("Approximate throughput: %.2f MB/s\n",
            throughput);
    }
}

return 0;
}

```

```

gowtham@DESKTOP-2F03K3T:~/OS$ gcc producer_consumer.c -o producer_consumer
gowtham@DESKTOP-2F03K3T:~/OS$ ./producer_consumer
Consumer received 128894 bytes
Producer sent 10000 messages
Communication time: 8188.00 microseconds
Approximate throughput: 12.21 MB/s
gowtham@DESKTOP-2F03K3T:~/OS$ █

```

2. Implement `ls -l | grep ".c"` Using `fork()`, `pipe()`, `dup2()`, and `exec()`

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t pid1, pid2;

    /* Create pipe */
    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    /* Create first child */
    pid1 = fork();

    if (pid1 < 0)
    {
        perror("fork");
        exit(1);
    }

    if (pid1 == 0)
    {
        /* Child 1 - ls */

        close(pipefd[0]);

        /* Redirect stdout to pipe */
        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("execlp");
        exit(1);
    }
}
```

```
/* Create second child */
pid2 = fork();

if (pid2 < 0)
{
    perror("fork");
    exit(1);
}

if (pid2 == 0)
{
    /* Child 2 - grep */

    close(pipefd[1]);

    /* Redirect stdin from pipe */
    dup2(pipefd[0], STDIN_FILENO);

    close(pipefd[0]);

    execlp("grep", "grep", ".c", NULL);

    perror("execlp");
    exit(1);
}

/* Parent */
close(pipefd[0]);
close(pipefd[1]);

waitpid(pid1, NULL, 0);
waitpid(pid2, NULL, 0);

return 0;
}
```

```

gowtham@DESKTOP-2F03K3T:~/OS$ nano ls_grep.c
gowtham@DESKTOP-2F03K3T:~/OS$ gcc ls_grep.c -o ls_grep
gowtham@DESKTOP-2F03K3T:~/OS$ ./ls_grep
-rwxr-xr-x 1 gowtham gowtham 16552 Aug  8 09:21 client
-rw-r--r-- 1 gowtham gowtham 1348 Aug  8 09:21 client.c
-rw-r--r-- 1 gowtham gowtham 226 Jul 28 03:43 file.c
-rw-r--r-- 1 gowtham gowtham 986 Aug 14 08:59 hello.c
-rw-r--r-- 1 gowtham gowtham 1737 Aug 14 09:38 keyboard_shell.c
-rw-r--r-- 1 gowtham gowtham 1738 Aug 14 09:29 keyboard_shell.c.save
-rw-r--r-- 1 gowtham gowtham 1179 Sep 12 08:18 ls_grep.c
-rw-r--r-- 1 gowtham gowtham 2559 Aug 29 10:08 memory.c
-rw-r--r-- 1 gowtham gowtham 3201 Aug 29 10:07 parser.c
-rwxr-xr-x 1 gowtham gowtham 16264 Aug  8 08:37 practical2
-rw-r--r-- 1 gowtham gowtham 1420 Aug  8 08:52 practical2.c
-rw-r--r-- 1 gowtham gowtham 1218 Aug  8 08:48 practical3.c
-rwxr-xr-x 1 gowtham gowtham 16344 Aug  8 08:56 practical4
-rw-r--r-- 1 gowtham gowtham 1787 Aug  8 08:59 practical4.c
-rwxr-xr-x 1 gowtham gowtham 16432 Aug  8 09:15 practical5
-rw-r--r-- 1 gowtham gowtham 1548 Aug  8 09:13 practical5.c
-rwxr-xr-x 1 gowtham gowtham 16296 Aug 14 09:05 process_demo
-rw-r--r-- 1 gowtham gowtham 1072 Aug 14 09:07 process_demo.c
-rwxr-xr-x 1 gowtham gowtham 16296 Aug 26 06:45 process_states
-rw-r--r-- 1 gowtham gowtham 2699 Aug 26 06:59 process_states.c
-rwxr-xr-x 1 gowtham gowtham 16464 Sep 12 08:13 producer_consumer
-rw-r--r-- 1 gowtham gowtham 1732 Sep 12 08:13 producer_consumer.c
-rw-r--r-- 1 gowtham gowtham 1728 Aug  8 09:20 server.c
-rw-r--r-- 1 gowtham gowtham 526 Aug 14 09:24 shell.c
-rw-r--r-- 1 gowtham gowtham 40 Aug  8 08:38 source.txt
-rw-r--r-- 1 gowtham gowtham 1802 Aug 29 09:48 tokenizer.c
-rw-r--r-- 1 gowtham gowtham 698 Aug 26 07:01 zombie.c
gowtham@DESKTOP-2F03K3T:~/OS$

```